

SIMATS ENGINEERING

Swetha.B
(192412215)

ITA0606
Machine learning

Experiment 3

3. Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier, plot_tree

from sklearn.metrics import accuracy_score

df = pd.read_csv("your_file.csv")

print("First 5 rows:")

print(df.head())

label_encoders = {}

for column in df.columns:

    le = LabelEncoder()

    df[column] = le.fit_transform(df[column].astype(str))

    label_encoders[column] = le

X = df.iloc[:, :-1]

y = df.iloc[:, -1]

X_train, X_test, y_train, y_test = train_test_split(

    X,

    y,

    test_size=0.2,

    random_state=42

)

model = DecisionTreeClassifier()
```

```

        criterion="entropy",
        random_state=42
    )
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    print("\nAccuracy:", accuracy_score(y_test, y_pred))
    plt.figure(figsize=(15,8))
    plot_tree(
        model,
        feature_names=X.columns,
        class_names=[str(i) for i in model.classes_],
        filled=True
    )
    plt.show()
    print("\nEnter values for a new sample:")
    new_sample = []
    for feature in X.columns:
        value = int(input(f"{feature}: "))
        new_sample.append(value)

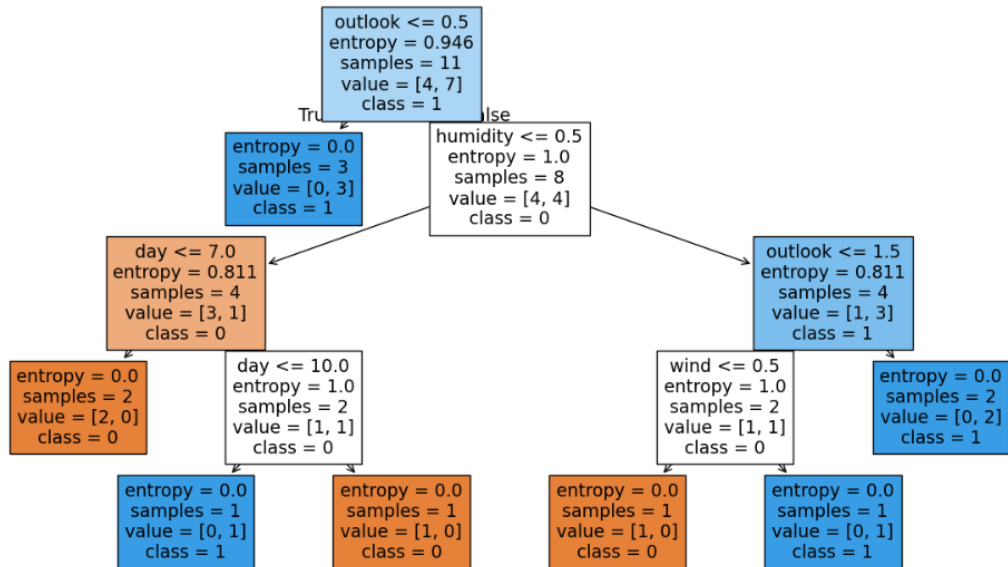
    prediction = model.predict([new_sample])
    target = df.columns[-1]
    predicted_label = label_encoders[target].inverse_transform(prediction)
    print("\nPredicted Class:", predicted_label[0])

```

Result:

```
... First 5 rows:  
day outlook temp humidity wind play  
0 D1 Sunny Hot High Weak No  
1 D2 Sunny Hot High Strong No  
2 D3 Overcast Hot High Weak Yes  
3 D4 Rain Mild High Weak Yes  
4 D5 Rain Cool Normal Weak Yes
```

Accuracy: 1.0



Enter values for a new sample: